

在坐标系A中的位置描述 ${}^A P = [{}^A p_x \quad {}^A p_y \quad {}^A p_z]^T$ B坐标系在A的位置 ${}^A B R = {}^B R^T = {}^B R^{-1}$ B坐标系的A的姿态描述 ${}^A B R = [{}^A \hat{X}_B \quad {}^A \hat{Y}_B \quad {}^A \hat{Z}_B] = [{}^B \hat{X}_A \quad {}^B \hat{Y}_A \quad {}^B \hat{Z}_A]^T = {}^B R^T$	速度变换/角速度变换 $\dot{Q}$ 是空间中动点。{A}和{B}是动坐标系，求 ${}^B \dot{V}_Q$ 与 ${}^B V_Q$ 的关 $\dot{V}_Q = \dot{V}_Q^{R^B V_{Coro}} = \dot{V}_Q^{R^B V_{Coro}}$	在连杆i上的惯性力和力矩： ${}^i F_i = m_i \cdot \ddot{v}_{C_i}$ ${}^i N_i = {}^i C_i \cdot {}^i \dot{\omega}_i + {}^i \omega_i \times {}^i C_i \cdot {}^i \omega_i$
$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R & {}^A P_{BORG} \\ 000 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$ ${}^B T_A = \begin{bmatrix} {}^B R & -{}^B R^A O_B \\ 0 & 1 \end{bmatrix}$ $R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} R_Y(\theta) = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix} R_Z(\theta) = \begin{bmatrix} c & -s & 0 \\ s & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_{Z^Y X^Z}(\gamma, \beta, \alpha) = R_z(\alpha) R_y(\beta) R_x(\gamma)$ $Z-Y-X\text{欧拉角} \quad \text{先绕 } \hat{Z}_B \text{ 转 } \alpha \text{ 角再 } \hat{Y}_B \text{ 绕转 } \beta \text{ 角再绕 } \hat{X}_B \text{ 转 } \gamma \text{ 角}$ $R_{Z^Y X^Z}(\alpha, \beta, \gamma) = R_z(\alpha) R_y(\beta) R_x(\gamma)$ $= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$ $= \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$ 欧拉角和固定角表示相同(旋转顺序不同) 命题： $R_z(\pm\pi+\alpha)R_y(\pm\pi-\beta)R_x(\pm\pi+\gamma) = R_z(\alpha)R_y(\beta)R_x(\gamma) \quad \alpha, \beta, \gamma \in [-\pi, \pi]$ 反解ZYX欧拉角 $-90.0^\circ \leq \beta \leq 90.0^\circ$ $\beta = \text{Atan2}(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}) \quad \alpha = \text{Atan2}(r_{21}/c\beta, r_{11}/c\beta)$ $\gamma = \text{Atan2}(r_{32}/c\beta, r_{33}/c\beta)$ $\beta = \pm 90.0^\circ \quad \text{仅能求出 } \alpha \text{ 和 } \gamma \text{ 的和或差} \quad \text{取 } \alpha = 0.0^\circ$ Z-Y-Z欧拉角 $R_{Z^Y Z^Z}(\alpha, \beta, \gamma) = R_z(\alpha) R_y(\beta) R_z(\gamma) \quad 0.0^\circ \leq \beta \leq 180.0^\circ$ $= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} c\gamma & -s\gamma & 0 \\ s\gamma & c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $= \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\beta c\gamma - s\alpha c\beta s\gamma & c\alpha c\beta s\gamma - s\alpha s\beta s\gamma & c\alpha c\beta \\ s\alpha c\beta c\gamma - s\alpha s\beta c\gamma - s\alpha c\beta s\gamma & s\alpha c\beta s\gamma - s\alpha s\beta s\gamma & s\alpha c\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}$ $\beta = \text{Atan2}(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}) \quad \gamma = \text{Atan2}(r_{32}/s\beta, -r_{31}/s\beta)$ $\beta = 0.0^\circ \text{ 或 } 180.0^\circ \quad \text{仅能求出 } \alpha \text{ 和 } \gamma \text{ 的和或差} \quad \text{取 } \alpha = 0.0^\circ$ 等效轴角表示 $V\varphi = 1 - c\varphi$ $R_K(\theta) = \begin{bmatrix} k_x^2 v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y^2 v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z^2 v\theta + c\theta \end{bmatrix}$ 反解等效轴角 规定 $\theta \in [0, \pi] \quad \theta = \text{Acos}\left(\frac{r_{11} + r_{22} + r_{33} - 1}{2}\right)$ 若 $\theta \in (0, \pi)$, $\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \frac{1}{2\sin\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \quad \text{唯一解}$ 右乘联体左乘基。 右乘T等效于先平移后旋转，左乘T等效于先旋转后平移	矢量和构造法求雅克比 $J(\Theta) = \begin{bmatrix} \dot{Z}_1 \times (P_N - P_1) & \dot{Z}_2 \times (P_N - P_2) & \dots & \dot{Z}_{N-1} \times (P_N - P_{N-1}) & 0 \\ \dot{Z}_1 & \dot{Z}_2 & \dots & \dot{Z}_{N-1} & \dot{Z}_N \end{bmatrix} \quad v_N = \begin{bmatrix} v_N \\ \omega_N \end{bmatrix} = J(\Theta)\dot{\Theta}$ 无冗余 $\dot{\Theta} = J^{-1}(\Theta)v_N$ 冗余 $\dot{\Theta} = J^T(JJ^T)^{-1}v_N$ 欠驱动 $\dot{\Theta} = (J^T J)^{-1}J^T v_N$ 奇异点 $\det(J(\Theta)) = 0 \quad \text{rank}(J(\Theta)) < m \quad \text{rank}(J(\Theta)) < n \quad (m, n)$ $\begin{bmatrix} \dot{r}_{11} & \dot{r}_{12} & \dot{r}_{13} \\ \dot{r}_{21} & \dot{r}_{22} & \dot{r}_{23} \\ \dot{r}_{31} & \dot{r}_{32} & \dot{r}_{33} \end{bmatrix} \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix}$ $\omega = \begin{bmatrix} 0 & -s\alpha & c\alpha s\beta \\ 0 & c\alpha & s\alpha s\beta \\ 1 & 0 & c\beta \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix} = B(\theta)\dot{\theta}$ $J(\Theta)\dot{\Theta} = \begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} \dot{d} \\ B(\theta)\dot{\theta} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & B(\theta) \end{bmatrix} \begin{bmatrix} \dot{d} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & B(\theta) \end{bmatrix} J_0(\Theta)\dot{\Theta} \quad J_0(\Theta)\dot{\Theta} = \begin{bmatrix} I & 0 \\ 0 & B^{-1}(\theta) \end{bmatrix} J(\Theta)\dot{\Theta}$ 关节驱动力矩 旋转关节 $\tau_i = {}^i n_i^T \cdot {}^i \hat{Z}_i$ 移动关节 $\tau_i = {}^i f_i^T \cdot {}^i \hat{Z}_i$ 力域中的雅可比 $\mathcal{F}$ 作用在末端执行器上 6×1 笛卡尔力-力矩矢量 $\mathcal{F}\delta\mathcal{X} = \tau\delta\theta$, 因为 $\delta\mathcal{X}\delta\theta$ 是 6×1 无穷小 $\mathcal{F}^T\delta\mathcal{X} = \tau^T\delta\theta$ 按照雅各比矩阵定义 $\frac{\delta\mathcal{X}}{\delta\theta} = J \frac{\delta\Theta}{\delta\theta}$ 则有 $\mathcal{F}^T\delta\mathcal{X} = \mathcal{F}^T J \delta\Theta = \tau^T\delta\Theta$ 最后得到 $\tau = J^T \mathcal{F}$	连杆i的势能可表示为： $u_i = -m_i \cdot g^T \cdot P_{C_i} + u_{ref_i}$ "g是3X1的重力矢量，"P _{C_i} 是连杆i质心的位置，u _{ref_i} 是使u _i 的最小值为零的常数 操作臂总势能：u = Σ _{i=1} ⁿ u _i $\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = \tau$ 操作臂运动方程： $\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = \tau$, τ是nX1驱动力矩矢量 状态空间方程： $\tau = M(\theta)\ddot{\theta} + V(\theta, \dot{\theta}) + G(\theta)$ 位形空间方程： $\tau = M(\theta)\ddot{\theta} + B(\theta)(\dot{\theta}\dot{\theta}) + C(\theta)(\dot{\theta}^2) + G(\theta)$ 此处，反对称性是指机械臂动力学方程 $M(\theta)\ddot{\theta} + V_m(\theta, \dot{\theta})\dot{\theta} + G(\theta) = \tau$ 中的质量矩阵M(θ)和矩阵V _m (θ, θ̇)之间的一个重要关系，即 矩阵Ṁ(θ) - 2V _m (θ,

